Assignment 2 — Snowflake Stream + Task Pipeline with GCS

Case Study: FreshMart Inventory Incremental Processing

FreshMart operates multiple warehouses. Its inventory application exports incremental inventory events as CSV files into Google Cloud Storage.

The company wants Snowflake to maintain two layers:

1. A raw inventory-event table containing every event loaded from GCS.
2. A current inventory table containing the latest stock position for each product and warehouse.

The operations team does not want developers to repeatedly process the complete raw table. Only new rows arriving after the previous processing cycle should be transformed.

Business Flow

GCS inventory files → Snowflake external stage → scheduled load → RAW inventory table → Stream → Task → CURRENT inventory table

Source File Structure

event_id

product_id

product_name

warehouse_id

qty_on_hand

unit_price

event_ts

inventory_001.csv

event_id,product_id,product_name,warehouse_id,qty_on_hand,unit_price,event_ts

1,P101,Wireless Mouse,W01,120,850,2026-08-12 08:00:00

2,P102,Keyboard,W01,75,1450,2026-08-12 08:01:00

3,P103,USB Hub,W02,60,1200,2026-08-12 08:02:00

4,P104,Webcam,W02,45,3100,2026-08-12 08:03:00

inventory_002.csv

event_id,product_id,product_name,warehouse_id,qty_on_hand,unit_price,event_ts

5,P101,Wireless Mouse,W01,105,850,2026-08-12 09:00:00

6,P102,Keyboard,W01,82,1450,2026-08-12 09:01:00

7,P105,Laptop Stand,W02,35,2100,2026-08-12 09:02:00

inventory_003.csv

event_id,product_id,product_name,warehouse_id,qty_on_hand,unit_price,event_ts

8,P103,USB Hub,W02,48,1200,2026-08-12 10:00:00

9,P104,Webcam,W02,38,3100,2026-08-12 10:01:00

10,P105,Laptop Stand,W02,50,2100,2026-08-12 10:02:00

Assignment Requirements

Build an incremental processing pipeline using GCS, Snowflake Streams and Snowflake Tasks.

Your implementation must demonstrate:

- Secure Snowflake access to the GCS bucket.
- An external stage for the inventory path.
- A raw inventory table.
- A current-inventory table where the business key is product + warehouse.
- A repeatable mechanism that loads new GCS files into the raw table.
- A Stream that exposes only unconsumed changes from the raw table.
- A Task that processes Stream data instead of scanning the full raw table every time.
- Existing product/warehouse rows are updated when a newer event arrives.
- New product/warehouse combinations are inserted.
- The Task avoids unnecessary transformation work when the Stream has no new data.
- Stream records are consumed after successful processing.
- The result after each file is demonstrated.

Expected Business Result

After inventory_001.csv:

P101/W01 = 120

P102/W01 = 75

P103/W02 = 60

P104/W02 = 45

After inventory_002.csv:

P101/W01 = 105

P102/W01 = 82

P103/W02 = 60

P104/W02 = 45

P105/W02 = 35

After inventory_003.csv:

P101/W01 = 105

P102/W01 = 82

P103/W02 = 48

P104/W02 = 38

P105/W02 = 50

Investigation Questions

Explain:

- What is the purpose of a Stream?
- Does a Stream physically copy the whole table?
- Why process the Stream instead of the complete RAW table?
- What happens to Stream records after successful DML consumption?
- Why check whether the Stream has data before transformation?
- What is the difference between the file-load Task and the CDC-processing Task?

Deliverables

Submit an architecture diagram, GCS screenshot, stage/file-format details, raw-table result, Stream output before processing, current-table output after each cycle, task status/history, and answers to the investigation questions.